

INTERNET & SAAS (ZIA)

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Internet & SaaS (ZIA)

Adding NSS Feeds

A Nanolog Streaming Service (NSS) feed specifies the data from the logs that the NSS sends to the security information and event management (SIEM) system. You can filter the data so that you send only the data you need to the SIEM, and you can add up to 16 NSS feeds for each [NSS server](#). ([Web](#) and [Firewall](#) logs are each limited to 8 feeds per NSS server to ensure optimal performance.) Each feed can have different filters and fields, and a different output format (e.g., CSV). To learn more about how to configure each feed, see:

- [Adding NSS Feeds for Web Logs](#)
- [Adding MCAS NSS Feeds](#)
- [Adding NSS Feeds for Firewall Logs](#)
- [Adding NSS Feeds for DNS Logs](#)
- [Adding NSS Feeds for Tunnel Logs](#)
- [Adding NSS Feeds for SaaS Security Logs](#)
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- [Adding NSS Feeds for Sandbox Verdict Logs](#)

Internet & SaaS (ZIA)

Adding NSS Feeds for Web Logs

You can configure up to 8 Nanolog Streaming Service (NSS) feeds that define the web logs that the NSS sends to the security information and event management (SIEM) system. You can also configure multiple filter types. For example, if an admin selects the location *HQ* and the department *Finance*, the NSS selects logs that belong to both the HQ location and Finance department. A large number of filters or complex filters, such as string searches, can impact the NSS performance.

Before you start configuring a feed for web logs, consider the [guidelines for configuring feeds](#).

To configure an NSS feed for web logs:

1. Go to Logs > Log Streaming > Internet Log Streaming - Nanolog Streaming Service.
2. On the NSS Feeds tab, click Add TCP NSS Feed.

The Add TCP NSS Feed window appears.

3. In the Add TCP NSS Feed window:
 - **Feed Name:** Enter or edit the name of the feed. Each feed is a connection between the NSS and your SIEM.
 - **NSS Type:** NSS for Web is selected by default.
 - **NSS Server:** Select an NSS server from the list. A [configured NSS server](#) is required to add an NSS feed.
 - **Status:** The NSS feed is Enabled by default. Choose Disabled if you want to activate it at a later time.
 - **SIEM Destination Type:** Select the type of destination:
 - **SIEM IP Address:** Appears when IP Address is selected as the destination type. Enter the IP address of the SIEM to which the logs are streamed.
 - **SIEM FQDN:** Appears when FQDN is selected as the destination type. Enter the destination for the TCP connection to which the logs are streamed. This allows failover from one IP to the other without manual intervention, but rather relying on updating the DNS entry. The NSS re-resolves the FQDN only when the existing connection goes down. This feature cannot be used for DNS-based load balancing.
 - **SIEM TCP Port:** Enter the port number of the SIEM to which the logs are streamed. Ensure that the SIEM is configured to accept the feed from the NSS.
 - **SIEM Rate:** Leave as Unlimited, unless you need to throttle the output stream due to licensing or other constraints.

- **SIEM Rate Limit (Events per Second):** This is only applicable if you selected **Limited** under **SIEM Rate**. Enter an appropriate rate limit for the events per second that you want streamed to your SIEM. This number must be between 100 and 1,000,000. A limit that is too low for the traffic volume causes log loss.
- **Log Type:** Choose **Web Log**.
- **Feed Output Type:** The output is a comma-separated (CSV) list by default. Choose **Tab-separated** to create a tab-separated list. Choose **Custom** to use a different delimiter, such as a dash, and enter the delimiter when you specify the feed output format. This menu also lists feed output types for specific SIEMs.
- **Feed Escape Character:** The Zscaler service hex encodes all non-printable ASCII characters that are in URLs when it sends logs to the NSS. Any URL character that is less than 0x21, or above 0x7E, is encoded as %HH. This ensures that your SIEM is able to parse the URLs in case they contain non-printable characters. For example, a \n char in a URL is encoded as %0A, and a space is encoded as %20. In this field, you can specify additional characters that you want to encode. For example, enter a comma (,) to encode it as %2C. This is useful if you are using this character as your delimiter and want to ensure it does not cause erroneous delimitation. Note that the service encodes characters in URLs, hostnames, and referrer URLs only. If custom encoding was done for a record, the %s{eedone} field is YES for that record.
- **Feed Output Format:** These are the fields that are displayed in the output. You can edit the default list and if you chose **Custom** as the **Feed Output Type**, change the delimiter as well. To learn more about the available fields and their syntax, see [NSS Feed Output Format: Web Logs](#).
- **Time Zone:** By default, this is set to the organization's time zone. The time zone you set applies to the time field in the output file. The time zone automatically adjusts to changes in daylight savings in the specific time zone. The configured time zone can be output to the logs as a separate field. The list of time zones is derived from the IANA Time Zone database. Direct GMT offsets can also be specified.
- **Duplicate Logs:** To ensure that no logs are skipped during downtime, specify the number of minutes that the NSS sends duplicate logs. Zscaler recommends setting the number to 5 minutes, or more if required. This allows the NSS to send up to 60 minutes (one hour) of logs to the SIEM after the connection is restored. To learn more, see [General Guidelines for NSS Feeds and Feed Formats](#).

4. Define the filters:

▼ Action

- **Policy Action:** Use this filter to limit the logs to transactions that were either allowed or blocked. Transactions wherein the service displayed a Caution page are considered

blocked transactions; if users proceeded with the transactions, they are considered allowed.

- **Policy Reason:** Use this filter to limit the logs based on the policy that the Zscaler service applied. These are the policy reason strings that are in transaction drilldown. They indicate which policy caused a block, or if allowed, the conditions under which they were allowed, such as **Allowed due to override** and **Internet Access cautioned**. Multiple selections are allowed.

▼ Who

- **Users:** Use this filter to limit the logs to specific users who generated transactions. You can search for users by username or email address. There is no limit on the number of users that you can select. Users that are deleted after they are selected appear with a strikethrough line.
- **Departments:** Use this filter to limit the logs to specific departments that generated transactions. You can search for departments. There is no limit on the number of departments that you can select. Departments that are deleted after they are selected appear with a strikethrough line.

▼ From Where

- **Locations:** Use this filter to limit the logs to specific [locations](#) from which transactions were generated. You can search for locations. There is no limit on the number of locations that you can select. Locations that are deleted after they are selected appear with a strikethrough line.
- **Location Groups:** Use this filter to limit the logs to specific [location groups](#) from which transactions were generated. You can search for location groups. Multiple selections are allowed.
- **Client IP Addresses:** Use this filter to limit the logs based on a client's private IP address. You can enter:
 - An IP address (e.g., 198.51.100.100)
 - A range of IP addresses (e.g., 192.0.2.1-192.0.2.10)
 - An IP address with a netmask (e.g., 203.0.113.0/24)

You can enter multiple entries. Press **Enter** after each entry, then click **Add Items**. For item lists, you can view up to 500 items on a page; filter the list by searching for a word, phrase, or number contained in an item; and remove all items from the list (**Remove All**) or only items from a specific page (**Remove Page**). If you select **Remove All** or **Remove Page**, a confirmation window appears.

- **Public IP Addresses:** Use this filter to limit the logs based on a client's public IP address. The internal IP address is available if traffic forwarding is forwarded to the service through a GRE or VPN tunnel or from the XFF header. If the internal IP address is not available, the value is the same as the client IP address. You can enter:

- An IP address (e.g., 198.51.100.100)
- A range of IP addresses (e.g., 192.0.2.1-192.0.2.10)
- An IP address with a netmask (e.g., 203.0.113.0/24)

You can enter multiple entries. Press **Enter** after each entry, then click **Add Items**. For item lists, you can view up to 500 items on a page; filter the list by searching for a word, phrase, or number contained in an item; and remove all items from the list (**Remove All**) or only items from a specific page (**Remove Page**). If you select **Remove All** or **Remove Page**, a confirmation window appears.

- **Traffic Forwarding:** Use this filter to limit the logs based on the traffic forwarding method used to send traffic to the Zscaler service. Choose one or more of the listed methods or choose **Any**.

▼ Transaction

- **Direction:** Use this filter to limit the logs to either inbound or outbound traffic.
- **User Agents:** Use this filter to limit the logs to transactions associated with the user-agent string that the browser included in its GET request. Choose from the list of predefined user-agent strings or enter custom user-agent strings. Multiple selections are allowed.
- **Custom User Agent Strings:** Use this filter to limit the logs to specific user-agent strings. A user-agent string contains browser and system information that the destination server can use to provide appropriate content.
- **Protocol Types:** Use this filter to limit the logs to specific protocols. Supported protocols are HTTP, HTTPS, and FTP. Multiple selections are allowed.
- **Request Methods:** Use this filter to limit the logs based on the HTTP request method obtained from the client request. Multiple selections are allowed.
- **Response Codes:** Use this filter to limit the logs based on the HTTP response code obtained from the server or generated by the Public Service Edge for Internet & SaaS (ZIA). Multiple selections are allowed.
- **Request Sizes:** Use this filter to limit the logs based on HTTP request size. Enter either a specific size or a range with a dash. By default, the service uses bytes, but you can also specify KB, MB, GB, or TB (e.g., 10KB-1MB, 200).

You can enter multiple entries. Press **Enter** after each entry, then click **Add Items**. For item lists, you can view up to 500 items on a page; filter the list by searching for a word, phrase, or number contained in an item; and remove all items from the list (**Remove All**) or only items from a specific page (**Remove Page**). If you select **Remove All** or **Remove Page**, a confirmation window appears.

- **Response Sizes:** Use this filter to limit the logs based on HTTP response size. Enter either a specific size or a range with a dash. By default, the service uses bytes, but you can also specify KB, MB, GB, or TB (e.g., **10KB-1MB**, **200**).

You can enter multiple entries. Press **Enter** after each entry, then click **Add Items**. For item lists, you can view up to 500 items on a page; filter the list by searching for a word, phrase, or number contained in an item; and remove all items from the list (**Remove All**) or only items from a specific page (**Remove Page**). If you select **Remove All** or **Remove Page**, a confirmation window appears.

- **Transaction Sizes:** Use this filter to limit the logs based on transaction size, which is the header and body request or response size, or the request and response size. Enter either a specific size or a range with a dash. By default, the service uses bytes, but you can also specify KB, MB, GB, or TB (e.g., **10KB-1MB**, **200**).

You can enter multiple entries. Press **Enter** after each entry, then click **Add Items**. For item lists, you can view up to 500 items on a page; filter the list by searching for a word, phrase, or number contained in an item; and remove all items from the list (**Remove All**) or only items from a specific page (**Remove Page**). If you select **Remove All** or **Remove Page**, a confirmation window appears.

- **Referer URLs:** Use this filter to limit the logs based on the Referer URL in the HTTP header. You can use wildcards based on the rules:
 - ***string** → Suffix matching match URLs ending with 'string'
 - **String*** → Prefix matching match URLs beginning with 'string'
 - ***string*** → Substring matching match URLs containing 'string'
 - **String** → Exact matching match URLs that are exactly 'string'

Multiple strings are allowed. Enter one string per line, then click **Add Items**. String search is not case-sensitive. For item lists, you can view up to 500 items on a page; filter the list by searching for a word, phrase, or number contained in an item; and remove all items from the list (**Remove All**) or only items from a specific page (**Remove Page**). If you select **Remove All** or **Remove Page**, a confirmation window appears.

▼ To Where

- **URL Filter Type:** Use this filter to limit the logs based on URLs in HTTP Requests. You can specify either a **Hostname** or the **Full URL**. You can use wildcards based on the rules:
 - **String** → Exact matching match URLs that are exactly 'string'
 - ***string*** → Substring matching match URLs containing 'string'
 - **String*** → Prefix matching match URLs beginning with 'string'
 - ***string** → Suffix matching match URLs ending with 'string'
- **Hostnames:** This is only applicable if you select **Hostname** under **URL Filter Type**. Use this filter to limit the logs based on specific hostnames. For item lists, you can view up to 500 items on a page; filter the list by searching for a word, phrase, or number contained in an item; and remove all items from the list (**Remove All**) or only items from a specific page (**Remove Page**). If you select **Remove All** or **Remove Page**, a confirmation window appears.
- **Full URLs:** This is only applicable if you select **Full URL** under **URL Filter Type**. Use this filter to limit the logs based on specific URLs. For item lists, you can view up to 500 items on a page; filter the list by searching for a word, phrase, or number contained in an item; and remove all items from the list (**Remove All**) or only items from a specific page (**Remove Page**). If you select **Remove All** or **Remove Page**, a confirmation window appears.
- **URL Classes:** Use this filter to limit the logs to specific [URL classes](#). Select those that you want to include. Multiple selections are allowed.
 - **Include:** Use this filter to include selected URL classes in the logs. This filter is enabled by default. The default value for this filter is **Any**.
 - **Exclude:** Use this filter to exclude selected URL classes from the logs. The default value for this filter is **None**.
- **URL Super Categories:** Use this filter to limit the logs to specific [URL super categories](#). Select those that you want to include. Multiple selections are allowed.
 - **Include:** Use this filter to include selected URL super categories in the logs. This filter is enabled by default. The default value for this filter is **Any**.
 - **Exclude:** Use this filter to exclude selected URL super categories from the logs. The default value for this filter is **None**.
- **URL Categories:** Use this filter to limit the logs to specific [URL categories](#). Select those that you want to include. Multiple selections are allowed.

- **Include:** Use this filter to include selected URL categories in the logs. This filter is enabled by default. The default value for this filter is **Any**.
- **Exclude:** Use this filter to exclude selected URL categories from the logs. The default value for this filter is **None**.



The **URL Categories** filter falls under the same filter class as the **Advanced Threats** filter. This means if you select at least one advanced threat, then you must also select the specific URL categories that you want included in the logs. When you select at least one advanced threat, the default **Any** option in the **URL Categories** filter does not apply. In this case, you must select all the options in the **URL Categories** filter to include any possible URL categories in the logs.

- **Server IP Addresses:** Use this filter to limit the logs based on the destination server's IP address. You can enter:
 - An IP address (e.g., 198.51.100.100)
 - A range of IP addresses (e.g., 192.0.2.1-192.0.2.10)
 - An IP address with a netmask (e.g., 203.0.113.0/24)

You can enter multiple entries. Press **Enter** after each entry, then click **Add Items**. For item lists, you can view up to 500 items on a page; filter the list by searching for a word, phrase, or number contained in an item; and remove all items from the list (**Remove All**) or only items from a specific page (**Remove Page**). If you select **Remove All** or **Remove Page**, a confirmation window appears.

- **Cloud Application Classes:** Use this filter to limit the logs to the selected [cloud application classes](#). Multiple selections are allowed.
- **Cloud Applications:** Use this filter to limit the logs to selected [cloud applications](#). Multiple selections are allowed.
 - **Include:** Use this filter to include selected cloud applications in the logs. This filter is enabled by default. The default value for this filter is **Any**.
 - **Exclude:** Use this filter to exclude selected cloud applications from the logs. The default value for this filter is **None**.



The Miscellaneous <Cloud Application Category> Apps (e.g., Miscellaneous Finance Apps) option in this filter represents all the newly added lesser-known predefined applications for the category (e.g., Finance). Use the Search function to view and select the available Miscellaneous <Cloud Application Category> Apps options. To view the list of supported applications for each category, see [Viewing Supported Cloud Applications](#).

- **Application Segment:** Use this filter to limit the logs to specific [application segments](#). The default option for this filter is **Any**.

▼ Security

- **Malware Classes:** Use this filter to limit the logs based on malware class or name. Multiple selections are allowed.
- **Malware Names:** Use this filter to limit the logs based on specific malware or viruses that were detected. You can specify multiple malware or virus names. Use the Search function to search for either. Use the guidelines:
 - *string → Suffix matching match malware names ending with 'string'
 - String* → Prefix matching match malware names beginning with 'string'
 - *string* → Substring matching match malware names containing 'string'
 - String → Exact matching match malware names that are exactly 'string'
 Multiple strings are allowed. Enter one string per line. String search is not case-sensitive.
- **Advanced Threats:** Use this filter to limit the logs based on the types of advanced threats that were detected. Multiple selections are allowed.



The **Advanced Threats** filter falls under the same filter class as the **URL Categories** filter. This means if you select at least one URL category, then you must also select the specific advanced threats that you want included in the logs. When you select at least one URL category, the default **Any** option in the **Advanced Threats** filter does not apply. In this case, you must select all the options in the **Advanced Threats** filter to include any possible advanced threats in the logs.

- **AI/ML Detection Source:** Use this filter to limit the logs based on the AI/ML detection source for the transaction. Multiple selections are allowed.
- **Threat Names:** Use this filter to limit the logs based on specific threats that were detected. You can specify multiple threat names. Use the Search function to search

with the guidelines:

- *string → Suffix matching match threat names ending with 'string'
- String* → Prefix matching match threat names beginning with 'string'
- *string* → Substring matching match threat names containing 'string'
- String → Exact matching match threat names that are exactly 'string'

Multiple strings are allowed. Enter one string per line. String search is not case-sensitive. For item lists, you can view up to 500 items on a page; filter the list by searching for a word, phrase, or number contained in an item; and remove all items from the list (**Remove All**) or only items from a specific page (**Remove Page**). If you select **Remove All** or **Remove Page**, a confirmation window appears.

- **Suspicious Content:** Use this filter to limit the logs based on the [Page Risk Index score](#) of a transaction. Enter either a single value or a range of values, between 0 and 100. Multiple values are allowed. Press **Enter** after each entry, then click **Add Items**. For item lists, you can view up to 500 items on a page; filter the list by searching for a word, phrase, or number contained in an item; and remove all items from the list (**Remove All**) or only items from a specific page (**Remove Page**). If you select **Remove All** or **Remove Page**, a confirmation window appears.

▼ File Type

- **Download File Type Category:** Use this filter to limit the logs based on the [category of the type of file downloaded](#) during the transaction. Multiple selections are allowed.
- **Download File Type:** Use this filter to limit the logs based on the [type of file downloaded](#) during the transaction. Multiple selections are allowed.
- **Upload File Type Category:** Use this filter to limit the logs based on the [category of the type of file uploaded](#) during the transaction. Multiple selections are allowed.
- **Upload File Type:** Use this filter to limit the logs based on the [type of file uploaded](#) during the transaction. Multiple selections are allowed.
- **Download User-Defined File Type:** Use this filter to limit the logs based on the [user-defined type of file downloaded](#) during the transaction. Multiple selections are allowed. To learn more, see [Configuring Custom File Types](#).
- **Upload User-Defined File Type:** Use this filter to limit the logs based on the [user-defined type of file uploaded](#) during the transaction. Multiple selections are allowed. To learn more, see [Configuring Custom File Types](#).
- **Unscannable Type:** Use this filter to limit the logs based on an unscannable file type. Multiple selections are allowed. The following options appear under this filter:
 - **Encrypted File:** Encrypted or password-protected (e.g., GZIP, PDF)

- Undetectable File: Unable to determine the file type, based on multiple methods
- Unscannable File: Unscannable (e.g., corrupt archive)

▼ DLP

- DLP Engines: Use this filter to limit the logs to transactions in which data leakage was detected based on specific [DLP engines](#). Multiple selections are allowed.
- DLP Dictionaries: Use this filter to limit the logs to transactions in which data leakage was detected based on specific [DLP dictionaries](#). Multiple selections are allowed.

5. Click **Save** and [activate the change](#).